AJEETKUMAR UKANDE

AI Engineer | Data Scientist | Generative AI & Agentic AI Specialist

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PROFESSIONAL SUMMARY

AI/ML and Data Science professional with 6 years of experience designing and deploying Machine Learning, Generative AI, NLP and Agentic AI solutions across Healthcare, Insurance, Banking & Finance, CPG and Manufacturing. Skilled in building end-to-end ML pipelines, Retrieval-Augmented Generation (RAG) systems, LLM-powered applications and scalable cloud-native AI architectures. Proficient in Google Cloud, Microsoft Azure and AWS, with a focus on solving complex business problems, automating workflows and delivering measurable business impact.

TECHNICAL SKILLS

- **Programming Languages**: Python, SQL
- **Machine Learning**: EDA, Feature Engineering & Selection, Linear & Logistic Regression, Ridge, Lasso, KNN, Decision Tree, Random Forest, SVM, AdaBoost, XGBoost, Model Evaluation & Hyperparameter Tuning
- **Deep Learning**: ANN, CNN, RNN, LSTM, GRU, Encoder-Decoder, Regularization, Dropout, Optimizers
- **NLP**: Tokenization, Stemming, Lemmatization, N-grams, TF-IDF, Word2Vec (CBOW), POS Tagging, NER, BERT
- **Generative AI & LLMs**: Transformers, Attention Mechanism, Hugging Face Pipelines, Prompt Engineering, RAG, Fine-Tuning, SFT, RLHF, Embedding Models, Conversation Memory | Models: GPT-5, GPT-4o, Gemini Pro / Flash, LLaMA 3.3 / 4, Mistral, Mixtral, Groq
- **Agentic AI**: LangChain, LangGraph, CrewAI, AutoGen, Agno, AI Agents, Multi-Agent Systems, Tool Calling, MCP, A2A
- **ML/DL Libraries**: Scikit-learn, TensorFlow, Keras, PyTorch, spaCy, NLTK, Pandas, NumPy, Matplotlib, Seaborn
- **APIs & App Frameworks**: FastAPI, Uvicorn, Flask, Pydantic, REST APIs, Postman, Swagger, Streamlit, Gradio, n8n
- **Databases**: MySQL, PostgreSQL, Snowflake, DynamoDB, Azure Cosmos DB
- **Vector Stores**: Azure AI Search, Pinecone, PGVector, ChromaDB, FAISS
- **Microsoft Azure**: Azure OpenAI, AI Foundry, Document Intelligence, Web Apps, Functions, ACR, Blob Storage / Data Lake Gen2, Key Vault, Application Insights, Auto-Scaling, Grafana
- **AWS**: SageMaker, Bedrock, Textract, Comprehend, Lambda, Glue, EC2, S3, ECR, IAM, Boto3
- **Google Cloud**: Vertex AI, Gemini API, Embeddings API
- **MLOps, LLMOps & DevOps**: Docker, GitHub Actions (CI/CD), Git, GitHub, Bitbucket, MLflow, LangSmith, Model Monitoring, Jira

Project

1. GenAI-Powered SDLC Assistant for Automated Jira Quality Check & Security Vulnerability Detection

- Designed and implemented a GenAI-powered solution to automate **Jira Story/Epic quality assessment, security impact analysis and risk classification**, reducing manual review effort for Information Security reviewers.
- Integrated **Jira APIs** to ingest Story/Epic fields including Summary, Description, Acceptance Criteria, tags and related metadata, followed by preprocessing and structured context construction for LLM analysis.
- Utilized **Google Gemini on Vertex AI** with prompt engineering and structured output techniques to generate story summaries, evaluate completeness and identify potential security-impact areas across domains such as **Identity & Access, Data/Cryptography, Application/Network and Vulnerability Management**. Implemented a rule-based + LLM-driven **risk scoring framework** to classify stories into risk categories and prioritize Major/Minor security-impacting changes for Information Security review.
- Developed automated decision logic to **auto-approve high-quality, low-risk stories** while routing ambiguous or security-impacting stories to human reviewers, maintaining a **human-in-the-loop** approval workflow. Built evaluation logic to measure **story completeness, classification accuracy and reviewer productivity**, helping achieve approximately **90% reduction in manual IS review effort**.
- Developed an interactive **Streamlit interface** and backend APIs using FastAPI for story analysis, risk visualization and reviewer decision support.

  Tech Stack: Python, Google Gemini, Vertex AI, LangChain, Jira REST API, FastAPI, Streamlit, Pandas, NumPy, Prompt Engineering, Structured Output, Risk Scoring, Human-in-the-Loop, R2D2

2. AI-Powered Attack Tree Generator – Cybersecurity Threat Modelling

- Designed and developed an AI-powered cybersecurity threat modelling platform using Google Gemini, Vertex AI, LangChain and Python to **automate attack tree generation, identify potential attack vectors, analyse threat scenarios, and recommend security mitigations**.
- Integrated **STRIDE-based threat modelling and Cyber Kill Chain** mapping to **systematically identify potential threats, attack paths, system vulnerabilities, and security risks, supporting security architecture reviews and risk assessment activities**.
- Automated the analysis of security assessment data from CSV and Excel files, reducing manual effort involved in identifying attack scenarios, mapping potential attack paths, and preparing threat modelling documentation.
- Developed a Streamlit-based application to generate visual attack tree diagrams using Mermaid and produce structured PDF and Markdown reports with threat scenarios, risk analysis, and actionable mitigation recommendations.
- Streamlined threat modelling and security assessment workflows by consolidating threat analysis, attack tree visualisation, and report generation into a single AI-assisted platform, enabling security teams to accelerate analysis and improve consistency in security reviews.
- **Tech Stack:** Python, Google Gemini, Google Cloud Vertex AI, LangChain, Streamlit, Pandas, STRIDE, Cyber Kill Chain, Mermaid, ReportLab.

3.Cyber Threat Intelligence Assistant using RAG Architecture

- Designed and developed an enterprise-grade Generative AI-powered Cyber Threat Intelligence Assistant using Google Cloud Vertex AI, Gemini, PostgreSQL, and PGVector to aggregate, index, and retrieve threat intelligence from multiple sources and security knowledge repositories.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline with semantic search, context-aware retrieval, and LLM-powered summarization to **help security teams quickly identify relevant threat indicators, attack patterns, emerging vulnerabilities, and actionable threat intelligence**.
- Automated manual threat intelligence research, cross-document information discovery, and security report generation, **reducing research and reporting time by approximately 80% and enabling security analysts to access relevant intelligence faster**.
- Developed context-grounded responses with source-relevant information to support threat investigation, security analysis, and faster decision-making while minimizing the effort required to manually search multiple threat feeds and documentation repositories.

Tech Stack: Python, Google Cloud Vertex AI, Google Gemini, RAG, PostgreSQL, PGVector, LangChain, Semantic Search, Vector Embeddings.

4. GenAI Agentic RAG-Based Chatbot for HVAC Product Maintenance

- Designed and implemented an enterprise Agentic RAG-based GenAI chatbot for HVAC service engineers, enabling intelligent troubleshooting, product maintenance guidance and technical knowledge retrieval across diverse enterprise data sources.
- Built an agentic orchestration layer using LangChain and LLMs to understand user intent, classify the query, determine the required reasoning path and dynamically decide whether to use RAG retrieval, an LLM/model, external APIs/tools or a direct response.
- Implemented a multi-model architecture supporting different LLMs/models based on task requirements, with Unstructured for processing diverse document formats and Azure Document Intelligence for document extraction. Integrated technical data from AWS S3 and Azure File Share and built preprocessing, chunking, metadata enrichment, and embedding pipelines.
- Developed an Agentic RAG workflow using Azure AI Search for semantic/vector retrieval, where the agent evaluates retrieved context and uses it as a tool for grounded response generation. Implemented conversation memory using Azure Cosmos DB to maintain relevant context across interactions and support multi-turn troubleshooting.
- Developed FastAPI-based backend services, containerized the solution using Docker and deployed it on Azure App Services. Implemented fallback and error-handling mechanisms across agents, models, retrieval and external tools, contributing to approximately 50% reduction in service resolution time.

Tech Stack: Python, Azure OpenAI, LangChain, Agentic RAG, Multi-Agent/Agentic Workflows, Multi-Model AI, LLM, Azure AI Search, Vector Search, Unstructured, Azure Document Intelligence, AWS S3, Azure File Share, Azure Cosmos DB, FastAPI, Docker, Azure App Services, Tool/API Integration, Prompt Engineering

5. Residential Fire Incident Classification Model (ML/NLP)

- Built and deployed an end-to-end **NLP/ML classification solution** to automate residential fire incident classification and streamline a multi-stage business review process. Built an AWS-based data pipeline using **AWS Sagemaker/Glue** to extract incident data from the Case Manager database and load curated datasets into **Snowflake and Amazon S3** for model development and analytics.
- Performed NLP preprocessing, feature engineering and model experimentation using Python, evaluating multiple **machine learning and deep learning classification models** for incident categorization. Developed reproducible training workflows using **Amazon SageMaker**, including preprocessing, model training, experiment tracking, evaluation and model selection. Containerized model workloads using **Docker** and registered validated models in **SageMaker Model Registry** to support controlled model versioning and deployment.
- Implemented model evaluation and experiment tracking to compare model combinations and select production-ready models based on defined performance criteria. Integrated model predictions with **MySQL and the AgilePoint frontend**, enabling business users to incorporate automated classification into the incident review workflow. Helped simplify the existing **three-stage incident review process to two stages**, reducing manual processing and contributing to approximately **70% improvement in business efficiency** while helping reduce exposure to potential legal penalties.
- **Tech Stack:** Python, NLP, Machine Learning, Deep Learning, TensorFlow, AWS SageMaker, SageMaker Model Registry, SageMaker Experiments, AWS Glue, Amazon S3, Snowflake, MySQL, Docker, MLOps

Tech Stack: Python, NLP, Machine Learning, Deep Learning, TensorFlow, AWS SageMaker, Glue, Snowflake, S3, MySQL, Docker, MLOps

Experience

Citi - AI Engineer (Gen AI, Agentic AI) ● 07/2025 - Present

- Design and develop enterprise **GenAI, Agentic AI, RAG, and Cybersecurity AI solutions** using Google Gemini, Vertex AI, LangChain and LangGraph. Build AI-driven platforms for **Jira story quality assessment, security risk analysis, threat intelligence, incident management and automated IT operations**.
- Collaborate with security architects, business stakeholders and engineering teams to translate business and security requirements into scalable AI solutions. Own solutions end-to-end, including **AI architecture, prompt engineering, agent workflows, API/tool integration, evaluation, deployment and production support**, with a focus on automation and measurable business impact.

Xoriant - Data Scientist (Gen AI, Agentic AI) ● 07/2024 - 07/2025

- Developed an AI-powered Attack Tree Generation platform leveraging Google Gemini, Vertex AI, LangChain and STRIDE framework to automate security architecture reviews, threat analysis and risk identification. Built an enterprise-grade RAG-based Threat Intelligence Research Assistant using PostgreSQL, PGVector, semantic search and Vertex AI embeddings, enabling conversational access to cybersecurity knowledge repositories.
- Built and deployed APIs and AI services with FastAPI and Docker, and worked with cross-functional teams to integrate them into enterprise platforms.
- Developed secure production-ready AI applications with authentication, monitoring, reporting automation, session management and governance controls. Collaborate with cross-functional teams to translate complex security and business requirements into actionable AI solutions, accelerating decision-making and improving operational efficiency. Mentor team members and promote AI/GenAI best practices, reusable frameworks and engineering standards across projects.

Tiger Analytics - Machine Learning Engineer ● 06/2022 - 06/2024

- Designed, built and productionized machine learning, deep learning and NLP solutions for predictive analytics and classification use cases.
- Performed data preprocessing, EDA, feature engineering, model training, validation and evaluation using Python and Scikit-learn.
- Built ETL and ML pipelines and handled deployment and monitoring on AWS (SageMaker, Glue), Snowflake and Docker.

- Supported model optimization, troubleshooting and continuous improvement, and collaborated with data science, engineering and business teams.

Grasp Technologies - ML Developer
● 06/2020 - 06/2022

- Developed machine learning models for classification, regression, clustering and predictive analytics.
- Performed data cleaning, preprocessing, feature engineering, hyperparameter tuning and model evaluation.
- Built reusable ML pipelines and Python data-processing and automation scripts.
- Prepared Excel and Power BI reports and dashboards to support business decisions.

EDUCATION

- Liverpool John Moores University (LJMU) – MSc in Data Science & Artificial Intelligence
- IIIT Bangalore – Executive Post Graduate Programme in Data Science & AI
- Pune University – Bachelor of Engineering (B.E.)

CERTIFICATIONS

- Microsoft Certified – Azure AI Engineer Associate